Analysis Choices and Ranking Stability in Kinase Inhibitor Selectivity Scores

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Abstract

Kinase selectivity scores can rank the same compounds differently. We compare four common scores across four profiling datasets. Agreement changes with affinity weighting, the compounds ranked and the treatment of pairs without detected binding. We relate ranking sensitivity to binding profile features. We propose four criteria for evaluating selectivity scores. No score is consistent across activity boundaries, so the boundary must be fixed by protocol. We construct a gated entropy that gives weight to weak binding and prove that adding a weakly bound target cannot make a compound appear more selective. On these data, it ranks compounds almost as the share of targets with detected binding does. Scores that follow this share recover full-panel rankings from fewer targets than the ratio and the published entropy, but on small panels many compounds with few active targets cannot be scored. These findings support reporting analysis choices, ranking sensitivity and compound coverage.

Keywords: binding selectivity, cancer drug discovery, cheminformatics, kinase inhibition, research methodology.

Introduction

Kinase inhibitors are among the most intensely investigated compound classes in drug discovery, with 94 FDA-approved agents and several hundred in clinical development [1]. The human kinome is comprised of more than 500 proteins with a conserved ATP-binding site [2], and selectivity is a primary criterion in lead optimization [3]. Profiling assays measure a compound against a panel of tens to hundreds of kinases, and the resulting profile is often reduced to one selectivity score. Several scores are in use:

- the S-score [4], the fraction of the panel bound below an affinity cutoff
- the selectivity entropy [5], the Shannon entropy of the profile treated as a distribution
- the Gini coefficient [6] of the profile
- the partition index [7], the share of binding that falls on one target
- ratio-based measures, which compare the primary target with an off-target.

They are often used interchangeably, although each summarizes a different aspect of the profile.

Earlier work established that these scores can disagree. Bosc et al. [8] compared six scores on the Davis and Anastassiadis datasets, computed several at more than one parameter value, compared them on compounds and kinases shared between datasets, and found that compound rankings differ substantially between them. They attributed individual disagreements to profile shape, such as a broad binder with one dominant target. The KInhibition platform [9] noted that conflicting results across metrics are common, and introduced a composite score. A later resource [10] from the same group offers a choice of scores. Miljkovic and Bajorath [11] found the concentration-dependent selectivity score of Klaeger et al. [12] broadly consistent with activity data from ChEMBL. Karaman et al. [4] and Uitdehaag et al. [13] showed that the S-score and the entropy of individual compounds vary when they are computed on random subpanels.

What has not been characterized systematically is how analysis choices alter agreement between the scores. Such choices include: how affinities are weighted, which compounds are ranked, how pairs without detected binding are treated, which assay targets count as kinases, and where each score's parameter is set. It is also not known which associations between profile shape and ranking instability persist across these choices, or how much of a ranking a smaller panel retains once some compounds can no longer be scored.

This work adds a systematic account of these choices. We compare the S-score, entropy, Gini and ratio on four datasets that span three assay technologies. Each is curated to one reference target per human protein kinase, with the entropy and Gini both in a common log-affinity form as well as in their published forms. We vary the compounds ranked, the treatment of pairs without detected binding and each score's parameter, and relate the resulting instability to features of the binding profile. From these results we state four criteria for a selectivity measure: activity eligibility, consistency across placements of the activity boundary, monotonicity when a weakly bound kinase is added, and orientation on profiles with equally strong targets. We evaluate the existing scores and simple comparators against them. We also give a gated, smooth-hinge entropy, prove that it satisfies all criteria. We then show that, once its coefficients are fixed, the criteria fix a unique measure. Stating such criteria follows the axiomatic approach Shannon[14] used for entropy. We measure how well each score's ranking is recovered from random subpanels, and for how many compounds, then give recommendations for reporting selectivity scores.

Methods

Datasets, Target Selection and Missing Values

Table 1 summarizes the four datasets. The Davis dataset[15] reports dissociation constants from a competition-binding assay, and the Klaeger dataset[12] apparent dissociation constants from Kinobeads chemoproteomics in cell lysate (beads carrying kinase inhibitors that capture

kinases from the lysate, with a mass-spectrometry readout). Both were converted, as in DeepDTA[16], to $\mathrm{p}K_d = -\log_{10}(K_d/10^9)$. The Anastassiadis dataset[17] reports residual kinase activity in a radiometric assay at 0.5 μM inhibitor, which we converted to percent inhibition (100 minus residual activity, clipped to $[0, 100]$). The Metz dataset[18] reports $\mathrm{p}K_i$ values from a medicinal-chemistry campaign. Because the assays and readouts differ, each score is computed on its dataset's own readout, and datasets are compared only through the rankings computed within each.

Table 1: The four datasets. The floor x_0 is the value given to pairs with no detected binding (Davis, Klaeger) or not tested (Metz); on Anastassiadis it is an operational cutoff for weak inhibition. τ^* is the activity threshold.

Dataset	Assay; readout	Compounds	Targets (released, curated)	x_0	τ^*
Davis	competition binding; $\mathrm{p}K_d$	68	433, 363	5.0	6
Klaeger	Kinobeads in lysate; $\mathrm{p}K_d$	222	343, 217	5.0	6
Anastassiadis	radiometric activity; % inhibition	178	300, 291	20%	50%
Metz	biochemical activity; $\mathrm{p}K_i$	704	172, 172	4.0	6

Target selection. Selectivity across kinases is expressed as a single value, so each kinase should be counted once. A panel that lists several constructs (recombinant forms of a protein, such as mutants or single domains) of one gene lists that kinase several times. The released Davis panel lists 433 constructs. They include mutant, phosphorylation-state and single-domain constructs of the same gene (12 for EGFR), 20 lipid kinases and three non-human kinases, and for 26 of the 68 compounds the two best-bound constructs are forms of one gene. We keep one reference construct per human protein-kinase gene: the wild-type construct, or its catalytic or N-terminal kinase domain where several domains are profiled. Genes profiled only as mutants are removed. The 343 Klaeger targets are all proteins that the beads captured and mass spectrometry measured, including complex subunits, such as cyclins, that bind the captured kinases. We keep the 217 single-gene protein kinases that Klaeger et al. annotate, except Q6ZSR9 (which UniProt lists as an uncharacterized protein without a kinase domain). This removes lipid kinases, four protein groups, and BCR and TRRAP, which

the source does not annotate as kinases and flags as possibly co-enriched with ABL1 and CDK8. On Anastassiadis we keep the first-listed of nine kinases profiled in two complexes or isoforms. The Metz columns are already one human protein kinase each. We call the targets kept on each dataset its *curated panel* (Table 1). The code repository lists every target with the reason it was kept or removed, and Supporting Information (Section S1) compares the released panels.

Floor-coded and missing values. On Davis, Klaeger and Metz, several kinds of pairs receive the same floor value x_0, which is not a measured detection limit. On Davis, the released matrix assigns $\mathrm{p}K_d = 5.0$ ($K_d = 10$ μM) to pairs with no detected binding, following DeepDTA[16]; Bosc et al.[8] used nearly the same value (10.1 μM). On Klaeger, we kept the 5,123 drug–target pairs that the original authors classified as high confidence. We coded at $\mathrm{p}K_d = 5.0$ three kinds of pairs: those with no detected binding, those whose fit was classified as low confidence, and the 194 measured affinities weaker than 10 μM. A low-confidence fit does not establish weak binding: of the 405 such pairs on curated targets, 375 have a fitted $\mathrm{p}K_d$ above 5. Giving all of these pairs the floor value is an analysis convention. On Metz, pairs that were not tested are coded at $\mathrm{p}K_i = 4.0$, the lowest value reported in the data. A median Metz compound is measured against only 21 of the 172 kinases, so we kept the 704 compounds measured against at least 50 (median 64). On Anastassiadis, the 1.1% of missing entries were set to 0% inhibition. Most pairs on this panel show little inhibition: 85% of the entries are at or below 20%, and each 5% interval above 20% holds less than 3% of the entries. We place x_0 at 20%, the upper end of these low values, as an operational cutoff for weak inhibition.

A compound is *active* against a target above the activity threshold τ^*: $\mathrm{p}K_d$ or $\mathrm{p}K_i$ 6 (K_d or K_i of 1 μM) on the affinity datasets, and 50% inhibition on Anastassiadis. With a Hill coefficient of 1, 50% inhibition at 0.5 μM inhibitor means an IC_{50} of 0.5 μM, close to the 1 μM of the other datasets. Supporting Information (Section S1) repeats the Davis

comparison of the scores with the threshold one log unit higher. An *active compound* has at least one target above τ^*.

Selectivity Scores and Their Implementations

A compound's profile is the vector $\mathbf{x} = (x_1, \ldots, x_N)$ of its readouts across the N panel targets, with higher values for tighter binding or stronger inhibition. Every score is oriented so that higher values are more selective (Table 2).

S-score. The S-score of Karaman et al.[4] is the fraction of the panel bound above a threshold,

$$S(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[x_i > \tau], \tag{1}$$

negated. They reported K_d cutoffs of 3 μM and 100 nM, which lie within our sweep of τ.

Selectivity entropy. The selectivity entropy of Uitdehaag and Zaman[5] is the Shannon entropy $H = -\sum_i p_i \log_2 p_i$ (in bits) of the profile normalized to a distribution, negated. In its published form, $p_i = K_{a,i} / \sum_j K_{a,j}$, with association constants $K_{a,i} = 1/K_{d,i}$ ($1/K_{i,i}$ on Metz). It needs affinity constants, so it is not computed on Anastassiadis. To place all four definitions on one scale, we also use a log-affinity form (on Anastassiadis, percent inhibition), which weights each target by its readout above a baseline β:

$$p_i = \frac{\max(x_i - \beta,\, 0)}{\sum_j \max(x_j - \beta,\, 0)}. \tag{2}$$

The baseline is a parameter we introduce; it has no published reference values.

Gini coefficient. Graczyk's Gini coefficient[6] measures how unequally binding is spread over the panel. Its published form uses percent-inhibition values. We compute it on the same

baseline-subtracted weights,

$$G(\beta) = \frac{2\sum_{i=1}^{N} i \cdot w_{(i)}}{N\sum_{i=1}^{N} w_{(i)}} - \frac{N+1}{N}, \qquad w_{(i)} = \max(x_i - \beta,\, 0) \text{ sorted ascending,} \tag{3}$$

and on Anastassiadis also in its native form.

Unless stated otherwise, both the entropy and Gini are in their log-affinity forms.

Ratio. The ratio compares the primary target with the k-th ranked off-target,

$$R_b(k) = x_{(1)} - \max(x_{(k+1)}, b), \tag{4}$$

where $x_{(j)}$ is the j-th largest value. The level b is the lowest coded value (5.0 on Davis and Klaeger, 4.0 on Metz and 0% on Anastassiadis), except in the recovery analysis, where it is x_0, and in D2, where it is swept. At this level $R_b(k)$ equals the ranking score of Bosc et al.[8] at rank $k+1$. We cap k at 5 because half of the Klaeger compounds have fewer than five active kinases.

Partition index. The partition index of Cheng et al.[7] is the fraction of a compound's molecules that would bind a reference target in a mixture of all panel kinases, $K_{a,\mathrm{ref}}/\sum_j K_{a,j}$. Only Klaeger names an intended target for each compound, so we use the most strongly bound target as the reference ($P_{\max}$), as Uitdehaag and Zaman[5] did. Like the published entropy, it is not computed on Anastassiadis. The CATDS score of Klaeger et al. is the same kind of fraction, for a designated target at one concentration. It needs Kinobeads dose–response curves, which the other three datasets lack, so it is not compared across datasets (Supporting Information, Section S6, compares a proxy).

Conventions. Where the entropy and Gini are undefined, we set both to 0. This ranks the compound as maximally selective under the entropy and minimally selective under the Gini. $R_b(k)$ uses b for a missing off-target and so scores every compound, as Bosc et al.'s

Table 2: Score implementations. Parameter ranges are for the pK_d datasets; on Anastassiadis τ runs over $[50, 90]\%$ and β over $[0, 30]\%$, and on Metz over $[5.0, 7.5]$ and $[4.0, 5.5]$. The last three rows are comparators.

Score	Weights or formula	Parameter swept	No target above the parameter
S-score	fraction above τ	$\tau \in [5.5, 8.0]$	score 0
Entropy, log-affinity	$\max(x_i - \beta, 0)$	$\beta \in [5.0, 6.5]$	undefined; set to 0 (top rank)
Gini, log-affinity	$\max(x_i - \beta, 0)$	$\beta \in [5.0, 6.5]$	undefined; set to 0 (bottom rank)
Ratio $R_b(k)$	$x_{(1)} - \max(x_{(k+1)}, b)$	$k = 1, \ldots, 5$	uses b
Entropy, published	$K_{a,i}$	none	—
Partition index $P_{\max}$	$\max_i K_{a,i} / \sum_j K_{a,j}$	none	—
Gated entropy	gated softplus (below)	none	not scored (D1)
Active count	number above τ^*	none	not scored (D1)
Entropy, $\beta = \tau^*$	$\max(x_i - \tau^*, 0)$	none	not scored (D1)
S-score, $\tau = x_0$	fraction above x_0	none	not scored (D1)

ranking score does. Log-affinity weights with $\beta = x_0$ give floor-coded pairs zero weight. Association constants give them a small positive weight if they are kept at the floor value. For the published entropy and $P_{\max}$, we give them zero weight. As a sensitivity analysis for the published entropy, we keep floor-coded pairs at the floor value. On Klaeger, we also give every measured high-confidence affinity its measured value, including the 194 weaker than the floor, and the other floor-coded pairs zero weight. No empirical data supports either assigning zero-weight or x_0 value to floor-coded pairs.

Comparators. The comparators are simple variants of existing scores. The only new score in this study is the gated entropy. The active count (the number of targets above τ^*, which is the numerator of the S-score at τ^*, with fewer more selective), the entropy with $\beta = \tau^*$ and the S-score with $\tau = x_0$ (the share of the panel with detected binding) are scored only for active compounds. Otherwise, a compound with no active target would have an active count of 0, fewer than any active compound, and would rank as the most selective.

Ranking Comparisons and Sensitivity Analysis

For each score and parameter value, compounds are ranked from most selective (rank 1), with ties given their average rank. Ranks are always computed among a stated set of compounds. Analyses of compounds with an active kinase rank them among themselves. Analyses that compare compounds with and without an active kinase rank all compounds. Agreement between two definitions is taken to be the Spearman correlation between their median rank vectors over each definition's parameter sweep: τ in steps of 0.25 (11 values) and β in steps of 0.25 (7 values) on the pK datasets, τ in steps of 5% (9 values) and β in steps of 5% (7 values) on Anastassiadis, and $k = 1, \ldots, 5$. Using mean ranks instead changes the correlations among all compounds by at most 0.08. Bootstrap 95% intervals use 1,000 resamples of compounds.

Each parameter value gives one ranking of the compounds. A compound's *rank instability* is the standard deviation of its ranks across one definition's sweep. Its *aggregated rank instability* covers all four definitions. Aggregation is done by taking the standard deviation of the compound's ranks over the rankings from all four sweeps together. Each definition therefore counts in proportion to its number of parameter values, and the S-score, with the most values, carries the most weight. We compute the Spearman correlation between within-definition instability and each profile feature: the number of active kinases, the top_1–top_2 affinity gap, the second-ranked affinity, and the standard deviation, range and skewness of the active affinities. A feature undefined for a compound with too few active kinases is set to 0. Supporting Information (Section S3) repeats the analysis with these compounds excluded. These associations are exploratory. A Bonferroni correction, checked against the Holm and Benjamini–Hochberg procedures, is applied only to the 24 tests per dataset in Supporting Information Table S4 (six features and four definitions), while other p-values are uncorrected.

To compare how much rankings change between datasets with how much they change between definitions, we matched 46 compounds profiled in both Davis and Klaeger by PubChem CID and restricted both panels to the 201 kinases they share (matched by gene symbol, with renamed symbols mapped to current ones). We then computed the Spearman

correlation between each definition's median ranking on Davis and on Klaeger, and between each pair of definitions' median rankings within each dataset.

Evaluation Criteria and the Gated Entropy

A selectivity measure f is a function that, for a given profile $\mathbf{x}$ of any length N, returns a real number $f(\mathbf{x})$, with higher values for more selective profiles. Let $B = [x_0, \tau^*]$ be the interval between the floor and the activity threshold. We present four criteria. D1 decides which compounds are scored, D2 and D3 concern how their scores change when the activity boundary moves and when the panel grows, and D4 concerns the direction of the score.

D1 (activity eligibility). $f(\mathbf{x})$ is undefined, or flagged as not eligible, whenever $\max_i x_i \leq \tau^*$. A compound with no target above τ^* has no target against which to judge selectivity. D1 is a reporting rule, so it is judged by construction: a score satisfies it if its definition withholds or flags these compounds. Bosc et al.[8] withheld ranks for such compounds under the S-score only, whereas D1 applies the rule to every score.

D2 (distributional consistency). Let θ be the parameter that places a score's activity boundary, i.e. the S-score's τ, the entropy's and Gini's β, the gated entropy's gate x_0 (with β moved with it and T fixed), and the ratio's level b. A pair of compounds *reverses in* B if $f(\mathbf{x};\theta) - f(\mathbf{y};\theta)$ is positive at some $\theta \in B$ and negative at another. D2 requires that at most $\varepsilon = 1\%$ of the pairs of compounds scored under D1 reverse in B, on every dataset and, for the ratio, for every k. D2 is therefore judged empirically, on the four datasets. The tolerance ε is our choice (Results report the range of ε over which the verdicts stay the same). To count reversals, each score is evaluated at a set of values of θ in B. The S-score changes only where τ passes an observed value, so it is evaluated at every observed value in B and its percentages are exact. The gated entropy jumps where its gate passes an observed value, so the gate is evaluated at each observed value in B, just below each, and at 201 equally spaced values. The other parameters are evaluated at the 201 values. Apart from the S-score's, the percentages are therefore estimates.

Some scores fix the boundary at one value, e.g. the active count, which counts the targets above τ^*. Such a score is assessed by sweeping that value over B, so that a fixed value does not let a score pass D2 vacuously. The value of θ is set in the analysis protocol according to the intended use, e.g. by whether binding weaker than τ^* matters at the concentration at which the compound will be used in practice. The resulting rankings are sensitive to this choice, and D2 reports this sensitivity as a diagnostic. At different values of θ, a score can measure different things: on Davis, the S-score at $\tau = 5$ counts the targets with K_d below 10 μM, and at $\tau = 6$ those below 1 μM. Its rankings can reasonably differ between the two.

D3 (monotonicity under weak-target addition). For any $\mathbf{x}$ with $\max_i x_i > \tau^*$, if $\mathbf{x}' = (x_1, \ldots, x_N, x_{N+1})$ with $x_0 < x_{N+1} \leq \tau^*$, then $f(\mathbf{x}') \leq f(\mathbf{x})$. Otherwise a compound can be made to look more selective by profiling it against more kinases it binds weakly. D3 is judged by construction: a score satisfies it if adding such a target can be proved never to raise the score, and violates it if a profile can be constructed on which it does. Tests on measured profiles (Results) are secondary checks of D3.

D4 (orientation). Fix the panel size $N \geq 2$ and all of the score's parameters. For every value a above τ^* and above every cutoff and baseline the score uses, let $\mathbf{v}_m$ be the profile with m targets at a and the other $N - m$ at the floor, for $m = 1, \ldots, N$. D4 requires $f(\mathbf{v}_{m+1}) \leq f(\mathbf{v}_m)$ for $m = 1, \ldots, N-1$, and $f(\mathbf{v}_1) > f(\mathbf{v}_N)$. These two conditions define what we take as the correct direction of a score: on the same panel, binding equally strongly to more targets must not make a compound look more selective, and binding to one target must be more selective than binding equally strongly to all. Strict decrease at every step is not required: $R_b(k)$ with $1 \leq k < N$ equals $a - \max(x_0, b)$ for $m \leq k$ and 0 for $m > k$. D4 is judged by construction.

When floor-coded pairs have zero weight, the profiles $\mathbf{v}_m$ become both broader and more evenly weighted as m increases. When they keep a positive weight, they form a background of small weights, and turning them into equally strong targets moves weight from this background into a few large weights. The distribution can then become less even and the

score rise, so a score can fail D4 because of its floor treatment, and each implementation is assessed with its own (example in Supporting Information, Section S4).

D4 constrains only the profiles $\mathbf{v}_m$. On a profile with unequal affinities, adding an active target that binds much more strongly than the others moves most of the weight to it, and a score that measures how concentrated binding is, such as the entropy or $P_{\max}$, then rates the compound as more selective (Supporting Information, Section S4). Whether that rating is correct depends on whether selectivity is taken to mean breadth or concentration, which D4 leaves open.

We state no criterion for the ratio's choice of reference off-target, because different references answer different questions.

The gated entropy. For a profile with $\max_i x_i > \tau^*$ (undefined otherwise, as D1 requires), define

$$w_i = \mathbf{1}[x_i > x_0]\; T \log\bigl(1 + e^{(x_i-\beta)/T}\bigr), \qquad p_i = \frac{w_i}{\sum_j w_j}, \qquad f(\mathbf{x}) = \sum_i p_i \log_2 p_i, \tag{5}$$

with the emphasis baseline $\beta = x_0$ and the smoothing width $T = \tau^* - x_0$ (1 pK unit on Davis and Klaeger, 2 on Metz, and 30 percentage points on Anastassiadis). It is the negative Shannon entropy of a gated, smoothed profile. The non-binder gate $\mathbf{1}[x_i > x_0]$ gives floor-coded pairs zero weight. The softplus replaces the entropy's hard cutoff $\max(x_i - \beta, 0)$, which gives a kinase just above β a weight close to zero.

Proposition 1 (monotonicity). With $\beta = x_0$, adding a kinase with a value in $(x_0, \tau^*]$ never raises f if and only if $T \geq (\tau^* - x_0)/\ln(2^e - 1) \approx 0.58\,(\tau^* - x_0)$. With $T = \tau^* - x_0$, it never raises f for any $\beta \in B$.

The proof of Proposition 1 (Supporting Information, Section S4) has two parts. For the *if* part, every gated target has weight above $m = T \ln 2$, and the added target has weight at most $e\,m$. This bound on its share of the total weight is enough for the entropy not to decrease. For the *only if* part, for any smaller T, a profile with one strong target and many

kinases just above the floor scores higher after a kinase is added at τ^*. For $\beta \neq x_0$ only the *if* direction is known.

Because a target at the floor has weight 0 and any target above it has weight above $m = T \ln 2$, the score jumps when a target's value crosses x_0. For example, on Davis or Klaeger ($x_0 = 5$, $T = 1$), take a profile with one active target at p$K_d = 7$. Adding a second target at the floor leaves the score at 0, and adding one just above the floor lowers it by 0.80 bits. Measurement noise near the floor could therefore change scores. To measure this effect, we perturbed the measured values above the floor (normal noise with standard deviation 0.1 and 0.3 pK units), set values pushed below the floor to the floor, and left floor-coded pairs fixed. On Anastassiadis, every entry was perturbed, with standard deviation 3 and 9 percentage points. Each setting used 50 draws. Noise can also change which compounds are scored: a compound whose strongest target lies just above τ^* can fall below it, and the reverse. D1 was therefore applied to each noisy profile, and rankings were compared over the compounds scored both with and without noise.

Proposition 2 (uniqueness for fixed weights). Fix the weights, that is, x_0, β and T in Equation (5), and let $\mathbf{p}$ be the shares of a profile's positive gated weights. Suppose a score has the form $-F(\mathbf{p})$, where F is continuous, increases with n when all n shares are equal, and has Shannon's grouping property[14]: when the kinases are split into groups, F equals its value across the groups plus the share-weighted values within them. Then the score is a positive multiple of the gated entropy and ranks compounds as the gated entropy does.

The proof applies Shannon's uniqueness theorem to the weight shares (Supporting Information, Section S4). Rényi entropies of order other than 1 lack the grouping property, so they are excluded.

Let us call shares $\mathbf{q}$ *more even* than shares $\mathbf{p}$ on the same panel if, for every j, the j largest entries of $\mathbf{q}$ sum to no more than the j largest entries of $\mathbf{p}$.

Proposition 3 (broadening). On a fixed panel, let a profile have positive weights $w_1, \ldots, w_r$ ($r \geq 1$) and zero weights elsewhere, and let a score be a function f of the profile's

weight shares with $f(\mathbf{q}) \leq f(\mathbf{p})$ whenever $\mathbf{q}$ is more even than $\mathbf{p}$. Changing one zero weight to a with $0 < a \leq \min_i w_i$ does not raise f.

The proof (Supporting Information, Section S4) shows that the added weight, being the smallest, makes the new shares more even than the old ones. The negative Shannon and Rényi entropies, the largest share and the Gini coefficient all have this property. For a score whose weight increases with affinity, such as the entropy, the gated entropy, $P_{\max}$ and the Gini, Proposition 3 therefore means that turning a floor-coded pair into a target bound no more strongly than the weakest weighted target never makes the compound look more selective. The condition on a is needed: adding a target tied with the strongest can lower the entropy.

Recovery from Smaller Panels

Profiling panels range from tens to hundreds of kinases, so a score computed on a smaller panel is only useful if it ranks compounds as it would on a larger one. We therefore measure how much of each score's full-panel ranking random smaller panels retain, and for how many compounds.

Subpanels of m targets were drawn uniformly at random from the curated targets, 50 per size, at $m = 10$, 20, 30, 40, 50, 65, 80, 100, 125, 150, 200, 250 and 300 below the full panel. Every score in Table 2, including the comparators, was computed on the same subpanels, with $\tau = \tau^*$ for the S-score, $\beta = x_0$ for the entropy and Gini, and $k = 1$ and $b = x_0$ for the ratio. D1 was applied to every score on every subpanel. *Coverage* is the share of compounds scored on the full panel that are still scored on a subpanel, averaged over the 50 subpanels of each size. It varies between subpanels, but on any one subpanel it is the same for every score as every score uses the same D1 rule (a compound is scored under all scores or none).

Recoverability is the share of clearly separated full-panel pairs (ranks at least a quarter of the scored compounds apart) that are scored on the subpanel in the same order, with a pair tied on the subpanel counted as one half. *Conditional agreement* is the Spearman correlation

between subpanel and full-panel scores over the compounds scored on both. Each score is compared with its own full-panel ranking, then, its clearly separated pairs are chosen from that ranking. Thus, recoverability measures how much of a score's own ordering a subpanel keeps. We write p^* for the smallest tested size at which the mean conditional agreement over subpanels exceeds 0.90, a descriptive benchmark that we chose. To test whether compounds whose score depends heavily on one target are harder to recover, we compute two quantities for each compound: its rank error at 50 targets, the mean absolute change in its rank (as a share of the compounds) over the subpanels on which it is scored, and the largest change in its full-panel score when one target is left out. We report the Spearman correlation between them over the scored compounds. As published, none of the four definitions has an eligibility rule, so Supporting Information (Section S5) also reports recovery with every compound scored, without D1. These results depend partly on the conventions for undefined scores. For example, the Gini recovers as fast as the entropy once the compounds with an undefined Gini which the convention ranks least selective are left out.

Results

Analysis Choices Change Agreement Between Scores

With the entropy and Gini in their log-affinity forms, and compounds with an active kinase ranked among themselves, the S-score, entropy and Gini agree more with one another than with the ratio (Table 3). On Davis, where every compound has an active kinase, they correlate at 0.963–0.999, and the ratio with each at 0.51–0.56. Among the 200 Klaeger compounds with an active kinase, they correlate at 0.89–0.997, and the ratio with each at 0.38–0.64. With log-affinity weights, the entropy and Gini, like the S-score, summarize how activity is spread over the panel, while the ratio instead compares the primary target with one off-target. That is, these two groups of measures calculate different things.

Table 3: Spearman correlations between median rankings. S: S-score; E: entropy; G: Gini; R: ratio, with the entropy and Gini in their log-affinity forms. E_{K_a}: the published entropy, with floor-coded pairs given zero weight and, in parentheses, kept at the floor. In the active columns, compounds are ranked among themselves. Every Davis compound has an active kinase.

	Davis	Klaeger	
Pair	all ($n = 68$)	all ($n = 222$)	active ($n = 200$)
S–E	0.963	0.896	0.897
S–G	0.963	0.508	0.892
E–G	0.999	0.658	0.997
S–R	0.563	0.078	0.378
E–R	0.513	0.344	0.622
G–R	0.510	0.655	0.635
E_{K_a}–S	0.738 (0.344)	0.571 (−0.597)	0.564 (−0.412)
E_{K_a}–G	0.736 (0.368)	0.530 (0.077)	0.782 (−0.094)
E_{K_a}–R	0.861 (0.688)	0.657 (0.648)	0.916 (0.591)

Weighting. The grouping of the S-score, entropy and Gini holds only for the log-affinity forms of the entropy and Gini. The published entropy, with association-constant weights, correlates more with the ratio than with the S-score or Gini on Davis, Klaeger and Metz, under both treatments of floor-coded pairs (Table 3; Supporting Information, Table S2). On Anastassiadis, the native percent-inhibition Gini correlates with the entropy at 0.63, and with the S-score and the ratio at only 0.34 and 0.33.

How far the published entropy is from the S-score and Gini depends on the treatment of floor-coded pairs. They make up 71% of the Davis matrix, and the published entropy's correlation with the S-score there is 0.34 with them at the floor and 0.74 with them given zero weight. The published and log-affinity entropies correlate at 0.37 (Davis) and −0.31 (Klaeger) with floor-coded pairs at the floor, and at 0.74 and 0.79 with them given zero weight (Figure 1). The negative Klaeger value comes from the floor-coded pairs, with which the published entropy also fails D4 ("Criteria and the Gated Entropy"). Using every measured Klaeger affinity at its measured value, including the 194 weaker than the floor, changes the published entropy's ranking little (Spearman 0.9996 with the floor-excluded ranking among

active compounds) and changes its correlations and recovery by at most 0.02 (Supporting Information, Section S2).

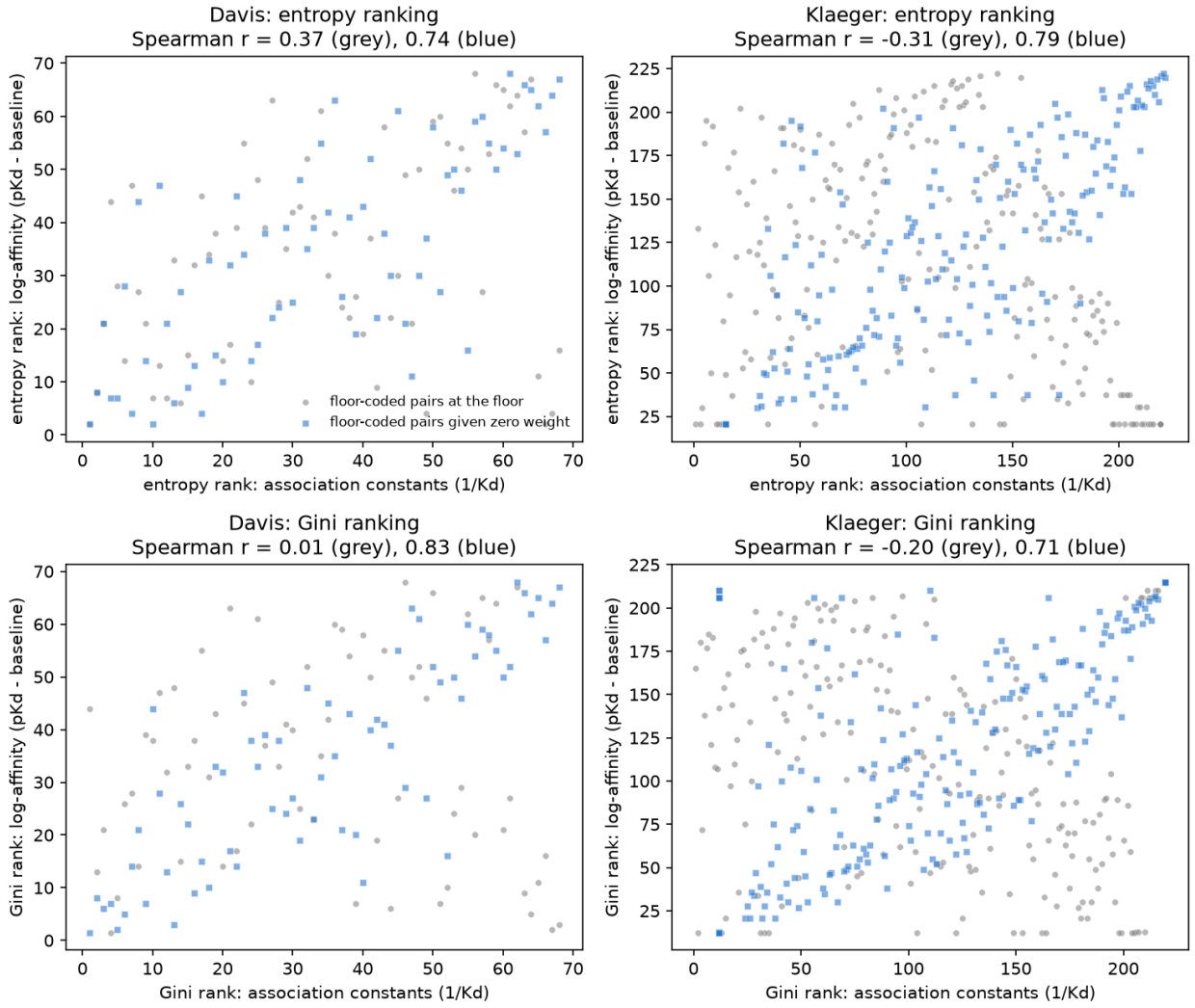


Figure 1: Rankings under association-constant weights (x-axis) against the baseline-subtracted log-affinity form (y-axis; median rank over the baseline sweep), on Davis and Klaeger. Top: entropy, for which association-constant weights are the published form. Bottom: Gini, shown for comparison; its published form uses percent inhibition. Grey: floor-coded pairs kept at the floor; blue: floor-coded pairs given zero weight. Spearman r above each panel.

Compounds ranked. Twenty-two Klaeger compounds have no target above the activity threshold. With them included, the Gini correlates more with the ratio (0.66) than with the S-score (0.51), although the difference is not significant (bootstrap 95% interval of the difference $[-0.35, 0.06]$, against $[0.16, 0.36]$ among active compounds). Each definition scores

these compounds by its own convention. The S-score and entropy rank these 22 compounds among the most selective (median ranks 24 and 20 of 222) and the ratio near the bottom (182). The Gini ranks them near the top where its score is defined and near the bottom where it is not (mean ranks 32 and 212). Their aggregated rank instability is $\sigma = 77.9$, against 32.2 for active compounds ($p < 0.001$, Mann–Whitney U test). Within a single definition's sweep, their rank instability is significantly higher than that of active compounds only for the Gini (65.0 against 16.3). The difference reverses for the S-score (19.0 against 25.7, $p = 0.01$), whose score is 0 for all of these compounds at thresholds of 6 or above. Sixteen of the 22 bind some kinase above the floor. Computed only over the configurations where their scores are defined, the mean instability of these 16 is lower than that of the active compounds within the entropy and the Gini (7.0 against 13.5, and 6.7 against 13.6; both $p = 0.002$). The other six show no binding to any curated target, although all are established inhibitors. For three of them, the target was quantified for other compounds (Supporting Information, Section S3). Whether a compound is eligible for a selectivity score therefore depends on the panel and the threshold. Supporting Information (Section S3) gives the per-definition contrasts.

Agreement within each dataset. Among compounds with an active kinase, the S-score, entropy and Gini correlate at 0.81–0.999 on all four datasets, and the ratio less with each (Supporting Information, Figure S1). On Anastassiadis, the separation is weak: the ratio correlates with the other three at 0.79–0.84, against 0.87–0.999 among them. With all compounds included, this falls to 0.48–0.55, because of the 45 compounds with no active kinase. On Metz, the ratio correlates with the other three at only 0.15–0.20, because the other three largely follow how many kinases each compound was tested against. Untested pairs are coded at the floor, and nearly every tested value (99.9%) lies above it. A compound tested against more kinases therefore has more kinases with positive weight and looks less selective under the S-score, entropy and Gini, while the ratio, which uses only the two highest values, is barely affected. The Spearman correlation between a compound's coverage and its

median rank (rank 1 = most selective) is 0.98 for the entropy, 0.95 for the Gini, 0.76 for the S-score and 0.11 for the ratio. Compounds measured against more kinases are also active against a larger share of those measured, but that association is much weaker ($r = 0.33$). The Metz correlations therefore show how coding untested pairs at the floor affects rankings.

Target selection and filtering. On the released Davis panel, the ratio agrees less with the other definitions than on the curated panel (0.34–0.46 against 0.51–0.56; Supporting Information, Section S1). Restoring the low-confidence Klaeger pairs shifts the all-compound Klaeger correlations by at most 0.08 and the active-compound correlations by at most 0.01. It leaves every p^* of the recovery analysis unchanged (Supporting Information, Section S1).

Effects of scoring against a different dataset versus different measure. On the 46 compounds profiled in both Davis and Klaeger, the same definition's Davis and Klaeger rankings correlate, on the 201 kinases both panels share, at 0.34–0.84 (mean 0.61), and different definitions' rankings within one dataset at −0.08–1.00 (mean 0.57). Among the 40 compounds active in both datasets, the means are 0.72 and 0.73, and the Gini's cross-dataset correlation rises from 0.45 to 0.84. On these compounds, the choice of definition moves rankings about as much as the change of dataset. The same compounds are active against 15.0% of the Davis panel and 3.9% of the Klaeger panel on average, and without restricting the panels to their shared kinases the cross-dataset correlations are lower (mean 0.53).

Parameter Sensitivity and Profile Associations

Moving each definition's activity boundary across B reorders a substantial share of compound pairs. The S-score's threshold reverses 8.4–13.0% of pairs of scored compounds on Davis, Klaeger and Anastassiadis, and 21.8% on Metz. The entropy's and Gini's baselines reverse 7.9–12.4% on the same three datasets, and 22.5–22.8% on Metz. The ratio's level b, which replaces any off-target value below b, measured or not, reverses 3.0–13.7% on Klaeger and 2.1–9.7% on Anastassiadis, rising with k, 0–2.6% on Davis and 0.04–1.03% on Metz. Sweeping

the gated entropy's gate x_0, with $\beta = x_0$ and T fixed at $\tau^* - x_0$, reverses 9.2–14.4% of pairs on Davis, Klaeger and Anastassiadis, and 23.6% on Metz. On Klaeger, the per-compound instability under the gate sweep matches that under the entropy's baseline sweep (mean rank $\sigma = 6.8$ and 7.0; Supporting Information, Figure S2). With the gate fixed, the gated entropy's remaining parameter, the emphasis baseline, reverses at most 0.72% of pairs. Supporting Information (Table S3) gives every value.

Reference off-target. When an active compound's two best-bound kinases have near-identical affinities, the ratio depends heavily on k. $R(1)$ compares the first kinase with the second, treating the second as an off-target (a kinase the compound is not meant to bind). For near-tied kinases this difference is close to zero, so the compound scores as non-selective. $R(2)$ compares the first kinase with the third, treating the second as a co-target (a second intended target), so the same compound can score as highly selective. On the 22 Klaeger compounds with an active kinase and a top_1–top_2 gap below 0.05, mean $R(1) = 0.02$ against $R(2) = 0.50$. Which reference is appropriate depends on whether the second kinase is an intended target, which the profile does not show. On Davis, fedratinib, a JAK2 inhibitor with JAK2 and GAK tied, has the most unstable ratio ranking (Figure 2, left).

Profile associations. Ranked among compounds with an active kinase, three associations between profile features and instability are significant at $p < 0.05$ on both pK_d datasets (Supporting Information, Table S4). The number of active kinases correlates negatively with entropy instability on every dataset ($r = -0.30$ to -0.42; on Davis only nominally, $p = 0.01$), and with Gini instability on Davis and Klaeger ($r = -0.31$ and -0.25). On Klaeger, entropy instability correlates with the share of a compound's above-floor kinases that lie within the baseline sweep ($r = 0.22$, $p = 0.002$). The top_1–top_2 gap correlates with ratio instability on Davis, Klaeger and Metz ($r = -0.44$, -0.28 and -0.26, all $p < 0.001$), but not on Anastassiadis ($r = 0.00$; -0.15, $p = 0.10$, without eight compounds whose two best targets are at the 100% inhibition ceiling). With a Bonferroni correction, only the

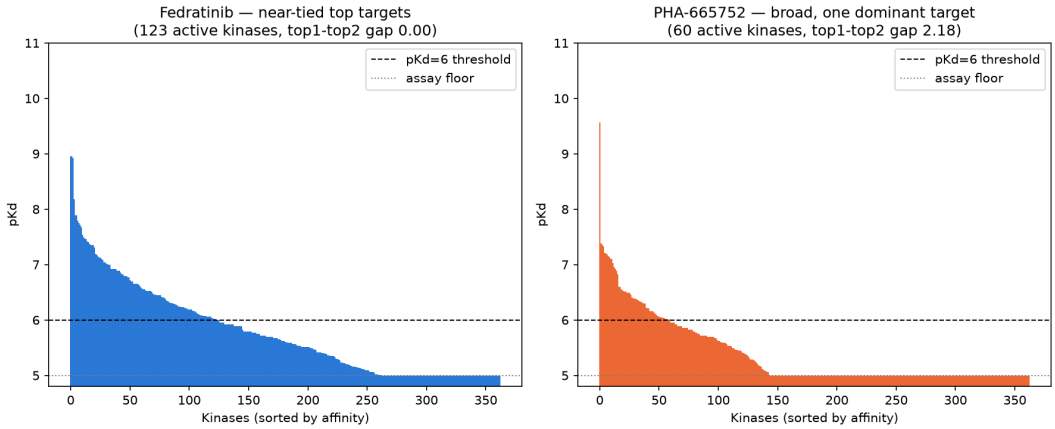


Figure 2: Binding profiles of two Davis compounds, targets sorted by affinity. Left: fedratinib, the most ratio-sensitive compound, with JAK2 and GAK tied among 123 active targets. Right: PHA-665752, the most unstable compound overall ($\sigma = 16.4$), with one dominant target among 60 active targets. Dashed line: p$K_d = 6$ activity threshold; dotted line: p$K_d = 5$ assay floor.

ratio–gap association remains significant on both pK_d datasets, as it does with the Holm and Benjamini–Hochberg procedures. On Klaeger (but not on Davis), S-score instability also correlates with the range of active affinities ($r = -0.32$; partial $r = -0.22$ with the number of active kinases held fixed). Ranked among all 222 Klaeger compounds, the skewness of the active affinities was the strongest correlate of Gini instability ($r = +0.26$). Ranked among active compounds, it is not significant ($r = +0.07$, $p = 0.31$), so the marginal association came mainly from the scoring of compounds with no active kinase. With the number of active kinases held fixed, a weaker association remains (Supporting Information, Section S3).

Broad profiles. Among active compounds, seven of the ten largest entropy–ratio disagreements on Klaeger involve profiles with 14–27 active kinases, and six of these have one clearly dominant target (top_1–top_2 gap above 1). The ratio ranks these seven 31st to 63rd of 200 and the entropy and Gini 146th to 180th. This is because the ratio reflects the dominant target, and the entropy and Gini reflect the number of active kinases. Bosc et al.[8] attributed disagreements for individual compounds to such profiles (i.e. broad profiles with one dominant target). On Davis, the most unstable compound overall is a profile of this kind (Figure 2,

right).

Criteria and the Gated Entropy

Table 4 gives each score's verdict on the four criteria, D1 to D4. D1, D3 and D4 are judged by construction, and D2 empirically on the four datasets (Methods). None of the four definitions, as published, withholds the score of a compound with no active kinase (so none meets D1), but any score can meet it by adopting that rule. The comparators and the gated entropy adopt it, i.e. they score only active compounds. No boundary parameter meets D2 on every dataset, and the verdicts are the same for any ε up to 13% (Supporting Information, Table S3). The published entropy and $P_{\max}$ have no boundary parameter and are not assessed against D2. D3 is the criterion on which the scores differ. The S-score with $\tau > x_0$, the entropy with $\beta = x_0$, the Gini and the published entropy violate it. The other scores, including $P_{\max}$, satisfy it. Every score in the table satisfies D4, and so do $P_{\max}$ and the published entropy with floor-coded pairs given zero weight.

Table 4: The four criteria for the four definitions, three comparators and the gated entropy. $P_{\max}$ and the published entropy have no boundary parameter and are assessed in the text. With $\beta < x_0$, the entropy and Gini fail D4. ✓: satisfied. ×: violated.

	S-score		Count	Entropy		Gini	Ratio	Gated
	$\tau > x_0$	$\tau = x_0$		$\beta = x_0$	$\beta = \tau^*$	$\beta \geq x_0$		entropy
D1 (eligibility)	×	✓	✓	×	✓	×	×	✓
D2 (consistency)	×	×	×	×	×	×	×	×
D3 (monotonicity)	×	✓	✓	×	✓	×	✓	✓
D4 (orientation)	✓	✓	✓	✓	✓	✓	✓	✓

By construction. These verdicts follow from each score's definition, without data. The S-score violates D3 for any threshold above the floor. That is, adding a kinase bound between the floor and the threshold increases N but not the number of hits, so the compound looks more selective. With the threshold at the floor, every addition that D3 allows is a hit, and a compound with C hits among N kinases moves from $-C/N$ to $-(C+1)/(N+1)$, which is not higher. The active count ranks compounds on any one panel as the S-score at τ^* does,

but it does not divide by N. A kinase added at or below τ^* is not counted, so the count and the score stay the same, and D3 holds. D3 therefore concerns how scores change when a panel is extended, not how compounds are ranked on one panel. The Gini violates D3, since a weakly bound kinase can raise its inequality. The ratio satisfies it, because a weak addition cannot displace the primary target or lower the reference off-target. $P_{\max}$ satisfies it too, since a weak addition leaves its numerator unchanged and raises its denominator. The entropy satisfies D3 at $\beta = \tau^*$, where weak binding carries zero weight. The gated entropy satisfies D1 through its eligibility rule and D3 by Proposition 1 in Methods. D3 concerns adding a target, not strengthening one already in the profile (so strengthening one is allowed to raise the gated entropy's score). Indeed, we observe that, if one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1, the score is -5.642. Raising one of the fifty to 6.0 raises it to -5.637.

D4 asks that a score not rise when a compound binds one more kinase as strongly as the others. For any compound that binds m kinases at the same value a, and the rest of the panel at the floor (such compounds are denoted by $\mathbf{v}_m$ in Methods), the scores are:

- active count: $-m$
- S-score with $\tau < a$: $-m/N$
- the entropies and the gated entropy, whose m weights are equal: $-\log_2 m$
- $P_{\max}$: $1/m$
- Gini: $(N - m)/N$
- ratio $R_b(k)$: $a - \max(x_0, b)$ for $m \leq k$ and 0 otherwise.

None of these values rises as m grows, so all of these scores satisfy D4. The values assume that floor-coded pairs have zero weight, which, for the entropy and Gini, requires $\beta \geq x_0$. On Anastassiadis, their sweeps also include baselines below the 20% floor, where both fail D4: on its 291 kinases, with $\beta = 0\%$ and targets at 60%, a second target raises the Gini from 0.0068 to 0.0135. With floor-coded pairs kept at the floor, the published entropy also fails

D4 on panels of the datasets' sizes: on 363 kinases with targets at $\mathrm{p}K_d = 7$, a second target raises its score from -7.41 to -6.76 bits. $P_{\max}$ satisfies D4 under either treatment, because its largest share falls as m grows.

D1 to D3 alone do not fix the direction of a score. The active count with its sign reversed, which calls a compound more selective the more active kinases it has, satisfies D1 and D3 because a kinase added at or below τ^* leaves it unchanged. Like the count, it fails D2, since reversing a score's sign does not change how many pairs reverse. It therefore has the same verdicts on D1 to D3 as the gated entropy, but it rises with the number of equally strong targets ($+m$ on $\mathbf{v}_m$), contradicting D4. The gated entropy without its non-binder gate also fails D4, because floor-coded pairs carry weight, and adding equally strong targets can concentrate the distribution. For the entropy, the gated entropy, $P_{\max}$ and the Gini, Proposition 3 allows us to make a conclusion even stronger than D4. On any profile (not only one with equally strong targets), turning a floor-coded pair into a target that is no stronger than the weakest weighted one does not raise the score.

By construction: counterexamples. The published entropy violates D3: with floor-coded pairs given zero weight, a profile with one target at $\mathrm{p}K_d = 6.01$ and fifty at 5.01 scores -5.35 bits, and adding one at 6.0 raises this to -5.19. The entropy violates D3 at $\beta < \tau^*$. Its hard cutoff gives a kinase just above β a weight close to zero. A profile with many such kinases therefore has high entropy. Adding one kinase bound more strongly, but still below τ^*, concentrates the distribution. With $\beta = 5.0$, a profile with one kinase at $\mathrm{p}K_d = 7.0$ and fifty at 5.1 scores -4.89 bits, and adding one at 6.0 raises this to -4.83. The gated entropy does not violate D3 on this profile: adding the kinase at 6.0 lowers its score from -5.642 to -5.666, because the softplus gives that kinase only 1.8 times the weight of a kinase at 5.1 (1.31 against 0.74). The entropy's hard cutoff gives it 10 times the weight (1.0 against 0.1).

Empirical checks. To additionally check empirically that Proposition 1 holds, we tested D3 for the entropy and the gated entropy on the measured profiles. On the measured profiles of the active compounds, adding a hypothetical kinase at τ^* never raises the entropy's score

on Davis, Klaeger or Metz at any baseline in their sweeps. On Anastassiadis, at baselines below the 20% floor (33% of the compounds at $\beta = 0\%$, 16% at 5%), it does. It never raises the gated entropy's score on any dataset. Adding measured kinases bound below the threshold to random 50- or 100-kinase subpanels of the same profiles raises the entropy's score in at most 0.1% of cases on any dataset, and never the gated entropy's.

On the four datasets, we compared the gated entropy's rankings with those of other scores, and tested their sensitivity to τ^* and to noise. On the data, the gated entropy's score falls as the number of active kinases rises (Spearman -0.78 to -0.97). Without the non-binder gate, the correlation reverses ($r = +0.91$ on Klaeger), as its failure of D4 predicts. It ranks the compounds almost exactly as the entropy does (0.994–0.998) and as the S-score with $\tau = x_0$ does (0.998–0.9997), which also satisfies D1, D3 and D4. Moving τ^* from 5.5 to 7 (40% to 70% on Anastassiadis) changes which compounds are scored, but does not change how those scored at both values are ranked (Spearman ≥ 0.999). With random noise added to the measured values, as described in Methods, D1 changed the status of at most 11 compounds in any draw. Over the compounds scored both with and without noise, the gated entropy's ranking correlated with its noise-free ranking at Spearman 0.978 or higher, and the entropy's at 0.983 or higher. The entropy satisfies D3 at $\beta = \tau^*$, where it gives zero weight to binding below the threshold. The gated entropy gives that binding a weight that varies smoothly with affinity and still satisfies D3, as proved in Methods. On these datasets, the gated entropy is thus not an empirical improvement on simpler scores.

Uniqueness. The criteria do not single out one score. No score satisfies D2. The active count, the S-score with $\tau = x_0$ and the entropy with $\beta = \tau^*$ satisfy D1, D3 and D4, as the gated entropy does. By Proposition 1, so does every gated entropy with $\beta = x_0$ and $T \geq 0.58\,(\tau^* - x_0)$. Once x_0, β and T are fixed, Proposition 2 makes the gated entropy unique: any score that depends only on the shares of its weights and meets Shannon's three conditions ranks compounds as it does. The criteria do not fix T, since any value above the bound passes, and variants of the gated entropy rank compounds almost as it does (Supporting

Information, Section S4).

Recovery from Small Panels

On small random panels, some compounds can no longer be scored. At 50 targets, 88% of the compounds scored on the full panel are still scored on Davis, 71% on Klaeger, 77% on Anastassiadis and 99% on Metz (Figure 3, top). The compounds lost are mostly those with few active targets: on a 50-target Klaeger subpanel, a compound with one active target stays scored with probability 0.23, and one with ten with probability 0.93. High conditional agreement on a small panel can therefore coexist with the loss of many narrowly active compounds. Coverage limits the recovery of pair order. Keeping 90% of clearly separated pairs takes 150 targets on Davis, 200 on Klaeger and Anastassiadis, and 40 on Metz for every score except the ratio, the published entropy and $P_{\max}$, which need as many or more (Figure 3, middle).

Among compounds scored on both panels, scores whose ranking depends mostly on the few strongest targets recover slowest (Figure 3, bottom). The S-score with $\tau = x_0$, the entropy with $\beta = x_0$, the Gini and the gated entropy rank compounds almost as the share of targets with detected binding does (Spearman above 0.97 with the S-score at $\tau = x_0$ on every dataset). We call these four the *share-following scores*. The curves of the other three differ from the gated entropy's by at most 0.015, so Figure 3 draws only the gated entropy. They recover fastest on Davis, Klaeger and Anastassiadis, reaching p^* at 20, 50 and 30 targets, against 30, 80 and 50 for the S-score at τ^* and the active count (which rank compounds identically on any one panel and share one curve in Figure 3), and 40, 100 and 65 for the entropy with $\beta = \tau^*$. On Metz, where untested targets are set to the floor, the S-score at τ^*, the active count and the entropy with $\beta = \tau^*$ recover fastest (50 targets, against 65 for the share-following scores). With bars of 0.80, 0.85 or 0.95 instead of 0.90, these fastest scores still reach the bar no later than any score outside their group. The ratio, the published entropy and $P_{\max}$ need far more targets. The published entropy and $P_{\max}$, with floor-coded pairs

given zero weight, need 300 of 363 targets on Davis and 200 of 217 on Klaeger. On Metz, the published entropy needs 150 of 172, while $P_{\max}$ needs more than the largest subpanel tested (150). The ratio needs 200 of 217 targets on Klaeger and more than the largest subpanel tested on the other datasets (300 of 363, 250 of 291 and 150 of 172). The published entropy gives weak binding a small positive weight, so including weak binding does not by itself make a score recover quickly. The gated entropy recovers rankings no better than the S-score (with $\tau = x_0$).

The value p^* is a threshold on the mean: at p^*, 62–90% of individual subpanels exceed 0.90 for the share-following scores (Supporting Information, Section S5). On Davis, Anastassiadis and Metz, compounds whose score changes most when one target is left out tend to have larger rank errors at 50 targets (Spearman $r = 0.21$–0.73 for the entropy with $\beta = x_0$, the Gini and the gated entropy). On Klaeger, this association is weak ($r \leq 0.14$).

Discussion

The scores compared here summarize different aspects of a binding profile, and the discrepancies between their rankings depend on choices made before any score is computed. The S-score, entropy and Gini agree more with one another than with the ratio when the entropy and Gini are computed from readouts above a common baseline and compounds without an active kinase are left out. With association-constant weights, or with compounds that have no active kinase included, this grouping weakens or disappears. When most entries are coded at the floor, as on Metz, the S-score, entropy and Gini agree for a different reason: each ranks compounds largely by how many kinases the compound was tested against. Bosc et al.'s window score correlates strongly with the ratio (0.91–0.96), and their ranking score is a ratio by definition. A proxy for the CATDS score of Klaeger et al., the top target's share of the profile, instead correlates more with the Gini than with the ratio on every dataset (Supporting Information, Section S6). The ratio and the S-score, entropy and Gini disagree most on

broad profiles with one dominant target. A selectivity ranking is therefore a statement about a score, its implementation, and a set of compounds and targets, not about the compounds alone. On compounds and kinases shared by two datasets, changing the definition moved rankings about as much as changing the dataset.

Reporting Recommendations

Each recommendation follows from a result above.

1. State which targets the score is computed over, including how the panel was curated, and the score implementation, including the weighting and any baseline.

2. Distinguish measured affinities, pairs with no detected binding, rejected fits and untested pairs, and state the value and weight each receives.

3. Apply D1: state the activity threshold, withhold or flag the scores of compounds with no target above it, and rank the eligible compounds among themselves. Otherwise, an active compound's instability depends on how the ineligible compounds are scored, as the Gini–skewness association showed.

4. Fix the activity boundary in the analysis protocol and report how rankings change when it is moved, for example by each compound's rank instability across the sweep. Rank instability is not a guide to robustness across datasets: on the 46 shared compounds, a compound's rank instability is not associated with how far its rank moves between Davis and Klaeger ($|r| \leq 0.18$; Supporting Information, Section S6).

5. For small panels, report how many compounds remain scorable as well as how well the ranking of those compounds is recovered.

6. When comparing studies that use different scores, fix the comparison protocol in advance and report the compounds on which the scores disagree.

Limitations

The four datasets, from three assay technologies, cover 68 to 704 compounds and 172 to 363 kinases, and the results may not hold for other compounds or panels. The datasets also differ in how many interactions they find: the 46 compounds profiled on both Davis and Klaeger are active against 15.0% of the Davis panel and 3.9% of the Klaeger panel on average. On Davis, Klaeger and Anastassiadis every compound was tested against the whole panel, and only 1.1% of the Anastassiadis entries are missing (set to 0% inhibition). On Metz, each compound was tested against some particular subset of the 172 kinases. On no dataset is the floor a measured detection limit. It codes non-detection (on Davis), non-detection, low-confidence fits and weak measured affinities (on Klaeger), and untested pairs (on Metz). On Anastassiadis, it is an operational cutoff for weak inhibition. Fixing the boundary at the floor therefore depends on how each dataset codes these pairs. Random subpanels are one sampling design, so panels chosen by kinase family, as many screening panels are, may recover rankings differently. Cell-free panels can also miss interactions that cellular target-engagement assays detect [19,20].

No score considered here uses target identity or exposure. Two compounds with the same affinity distribution receive the same score whether their off-targets are inert or linked to a class toxicity, and an affinity alone does not establish whether a target is occupied at a tolerated dose. A selectivity score is at best a summary of the binding profile, to be weighed with target-specific liability knowledge and exposure modeling. Selectivity rankings of the approved Klaeger compounds were not associated with the proportions of their adverse-event reports that were serious, fatal or led to hospitalization (Supporting Information, Section S7).

The four criteria play different roles. D1 is a reporting rule that any score can adopt. D2 is a diagnostic: no boundary parameter meets it, so the boundary must be fixed by protocol. D3 is the criterion on which the scores differ, and the gated entropy's guarantee covers the usual way a panel grows: most kinases added to a panel bind the compound weakly or not at all, as 83.3% of the Davis entries and 95.5% of the Klaeger entries lie below the activity

threshold. The guarantee does not extend to strengthening a target already in the profile. D4 calibrates the direction of a score on profiles with equally strong targets. Without it, D1 to D3 would not rule out the active count with its sign reversed. All four are proposed evaluation criteria and do not single out one score. On these datasets, the S-score with $\tau = x_0$ matches the gated entropy on every criterion and in recovery, and the entropy with $\beta = x_0$ ranks compounds almost identically, although it violates D3.

How closely a score follows the number of active kinases on real data, such as the gated entropy's Spearman -0.78 to -0.97, measures its agreement with breadth on those data, not whether it is correctly oriented. A score that measures concentration can correctly rank a compound with more active targets as more selective when its binding is more concentrated.

Conclusion

Kinase inhibitor selectivity rankings depend on choices usually left implicit: how affinities are weighted, which compounds are ranked, how pairs without detected binding are treated, which targets the panel includes, and where the activity boundary is placed. These choices change whether the common scores agree, how much a ranking moves when the activity boundary moves, and how much of a ranking a small panel retains. Selectivity analyses should state them, apply an activity-eligibility rule, fix the activity boundary in advance and report its effect, and report how many compounds remain scorable on small panels. The gated entropy shows that a Shannon entropy can give weight to binding below the activity threshold and remain monotone under weak-target addition, and that, once its parameters are fixed, Shannon's conditions make it unique, but on these data it ranks compounds almost as the share of targets with detected binding does.

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Supporting Information

Sections S1–S8 of the Supporting Information, with Tables S1–S9 and Figures S1–S4, give the full results for each analysis, the proofs of Propositions 1–3 and the script behind every reported number.

Data and Software Availability

All data and software used in this study are publicly available. The Davis kinase inhibitor profiling dataset is available at `https://github.com/dingyan20/Davis-Dataset-for-DTA-Prediction`. The Klaeger et al. dataset is available as supplementary data to the original publication (Science, 2017, `https://doi.org/10.1126/science.aan4368`). The Anastassiadis et al. (*Nat. Biotechnol.* 2011) and Metz et al. (*Nat. Chem. Biol.* 2011) datasets are available as supplementary data to their original publications (`https://doi.org/10.1038/nbt.2017`; `https://doi.org/10.1038/nchembio.530`). All analysis scripts, processed data files, and

the LaTeX source for this manuscript are available at `https://github.com/polinavino/kinase-selectivity-definitions`; this repository includes the code and full per-definition results for the secondary clinical-outcome analysis (FAERS adverse-event report rates for the FDA-approved Klaeger compounds, obtained via openFDA queries). The analyses used Python 3 with NumPy[21], pandas[22] and SciPy[23], and Supporting Information (Table S9) names the script behind every result. No confidential data were used in this study. All results reported in this manuscript are fully reproducible from the publicly available data and code. A preprint of this manuscript is available on ChemRxiv at `https://doi.org/10.26434/chemrxiv.15001618/v1`.

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Conflict of Interest

The author declares no conflict of interest.

References

[1] Roskoski, R. Properties of FDA-approved small molecule protein kinase inhibitors: a 2026 update. *Pharmacol. Res.* **2026**, *224*, 108107.

[2] Manning, G.; Whyte, D. B.; Martinez, R.; Hunter, T.; Sudarsanam, S. The protein kinase complement of the human genome. *Science* **2002**, *298*, 1912–1934.

[3] Bhullar, K. S.; Lagarón, N. O.; McGowan, E. M.; Parmar, I.; Jha, A.; Hubbard, B. P.;

Rupasinghe, H. P. V. Kinase-targeted cancer therapies: progress, challenges and future directions. *Mol. Cancer* **2018**, *17*, 48.

[4] Karaman, M. W.; Herrgard, S.; Treiber, D. K.; et al. A quantitative analysis of kinase inhibitor selectivity. *Nat. Biotechnol.* **2008**, *26*, 127–132.

[5] Uitdehaag, J. C. M.; Zaman, G. J. R. A theoretical entropy score as a single value to express inhibitor selectivity. *BMC Bioinformatics* **2011**, *12*, 94.

[6] Graczyk, P. P. Gini coefficient: a new way to express selectivity of kinase inhibitors against a family of kinases. *J. Med. Chem.* **2007**, *50*, 5773–5779.

[7] Cheng, A. C.; Eksterowicz, J.; Geuns-Meyer, S.; Sun, Y. Analysis of kinase inhibitor selectivity using a thermodynamics-based partition index. *J. Med. Chem.* **2010**, *53*, 4502–4510.

[8] Bosc, N.; Meyer, C.; Bonnet, P. The use of novel selectivity metrics in kinase research. *BMC Bioinformatics* **2017**, *18*, 17.

[9] Bello, T.; Gujral, T. S. KInhibition: a kinase inhibitor selection portal. *iScience* **2018**, *8*, 49–53.

[10] Saifudeen, M.; Zhu, S.; Liang, S.; et al. Comprehensive profiling of clinically approved kinase inhibitors reveals mutation-specific inhibitors and opportunities for drug repurposing. *Nat. Biotechnol.* **2026**, DOI: 10.1038/s41587-026-03090-8.

[11] Miljkovic, F.; Bajorath, J. Reconciling selectivity trends from a comprehensive kinase inhibitor profiling campaign with known activity data. *ACS Omega* **2018**, *3*, 3113–3119.

[12] Klaeger, S.; Heinzlmeir, S.; Wilhelm, M.; et al. The target landscape of clinical kinase drugs. *Science* **2017**, *358*, eaan4368.

[13] Uitdehaag, J. C. M.; Verkaar, F.; Alwan, H.; de Man, J.; Buijsman, R. C.; Zaman, G. J. R. A guide to picking the most selective kinase inhibitor tool compounds for pharmacological validation of drug targets. *Br. J. Pharmacol.* **2012**, *166*, 858–876.

[14] Shannon, C. E. A mathematical theory of communication. *Bell Syst. Tech. J.* **1948**, *27*, 379–423.

[15] Davis, M. I.; Hunt, J. P.; Herrgard, S.; et al. Comprehensive analysis of kinase inhibitor selectivity. *Nat. Biotechnol.* **2011**, *29*, 1046–1051.

[16] Öztürk, H.; Özgür, A.; Ozkirimli, E. DeepDTA: deep drug–target binding affinity prediction. *Bioinformatics* **2018**, *34*, i821–i829.

[17] Anastassiadis, T.; Deacon, S. W.; Devarajan, K.; Ma, H.; Peterson, J. R. Comprehensive assay of kinase catalytic activity reveals features of kinase inhibitor selectivity. *Nat. Biotechnol.* **2011**, *29*, 1039–1045.

[18] Metz, J. T.; Johnson, E. F.; Soni, N. B.; Merta, P. J.; Kifle, L.; Hajduk, P. J. Navigating the kinome. *Nat. Chem. Biol.* **2011**, *7*, 200–202.

[19] Binder, M. J.; Bashore, F. M.; Dunn Hoffman, K. K.; et al. Cellular context influences kinase inhibitor selectivity. *J. Med. Chem.* **2026**, *69*, 4208–4220.

[20] Binder, M. J.; Bashore, F. M.; Dunn Hoffman, K. K.; et al. Correction to "Cellular context influences kinase inhibitor selectivity". *J. Med. Chem.* **2026**, *69*, 20069–20071.

[21] Harris, C. R.; Millman, K. J.; van der Walt, S. J.; et al. Array programming with NumPy. *Nature* **2020**, *585*, 357–362.

[22] McKinney, W. Data structures for statistical computing in Python. Proceedings of the 9th Python in Science Conference. 2010; pp 56–61.

[23] Virtanen, P.; Gommers, R.; Oliphant, T. E.; et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **2020**, *17*, 261–272.

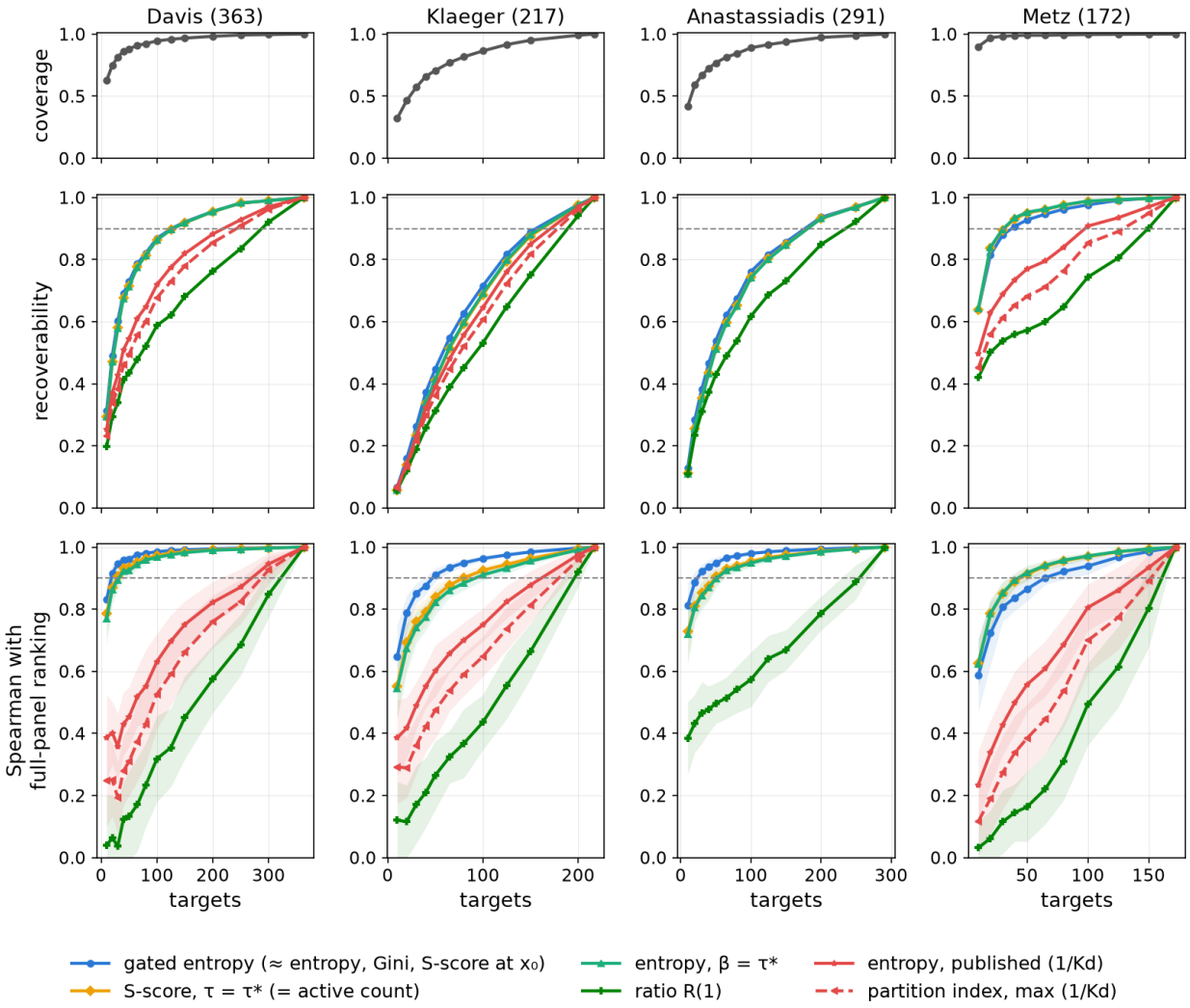


Figure 3: Recovery from random subpanels with D1 applied to every score, on all four datasets (50 subpanels per size). Top: coverage, the share of compounds scored on the full panel that are still scored. Middle: recoverability, the share of clearly separated full-panel pairs that are scored on the subpanel and keep their order. Bottom: mean Spearman correlation between subpanel and full-panel rankings over compounds scored on both, with ±1 standard deviation across subpanels shaded. Each score is compared with its own full-panel ranking. The published entropy and $P_{\max}$ give floor-coded pairs zero weight and are not defined on Anastassiadis. Dashed line: 0.90.